

OpenStax Calculus Volume 3

Section 7.3 Exercises: Applications

Source: <https://openstax.org/books/calculus-volume-3/pages/7-3-applications>. Exercises reproduced from OpenStax, licensed under CC BY 4.0.

This document was converted programmatically from OpenStax content; formulas were translated automatically from MathML and may contain conversion errors — verify against the source before relying on them.

1. A mass weighing 4 lb stretches a spring 8 in. Find the equation of motion if the spring is released from the equilibrium position with a downward velocity of 12 ft/sec. What is the period and frequency of the motion?
2. A mass weighing 2 lb stretches a spring 2 ft. Find the equation of motion if the spring is released from 2 in. below the equilibrium position with an upward velocity of 8 ft/sec. What is the period and frequency of the motion?
3. A 100-g mass stretches a spring 0.1 m. Find the equation of motion of the mass if it is released from rest from a position 20 cm below the equilibrium position. What is the frequency of this motion?
4. A 400-g mass stretches a spring 5 cm. Find the equation of motion of the mass if it is released from rest from a position 15 cm below the equilibrium position. What is the frequency of this motion?
5. A block has a mass of 9 kg and is attached to a vertical spring with a spring constant of 0.25 N/m. The block is stretched 0.75 m below its equilibrium position and released.
 - (a) Find the position function $x(t)$ of the block.
 - (b) Find the period and frequency of the vibration.
 - (c) Sketch a graph of $x(t)$.
 - (d) At what time does the block first pass through the equilibrium position?
6. A block has a mass of 5 kg and is attached to a vertical spring with a spring constant of 20 N/m. The block is released from the equilibrium position with a downward velocity of 10 m/sec.
 - (a) Find the position function $x(t)$ of the block.
 - (b) Find the period and frequency of the vibration.
 - (c) Sketch a graph of $x(t)$.
 - (d) At what time does the block first pass through the equilibrium position?
7. A 1-kg mass is attached to a vertical spring with a spring constant of 21 N/m. The resistance in the spring-mass system is equal to 10 times the instantaneous velocity of the mass.

- (a) Find the equation of motion if the mass is released from a position 2 m below its equilibrium position with a downward velocity of 2 m/sec.
 - (b) Graph the solution and determine whether the motion is overdamped, critically damped, or underdamped.
8. An 800-lb weight (25 slugs) is attached to a vertical spring with a spring constant of 226 lb/ft. The system is immersed in a medium that imparts a damping force equal to 10 times the instantaneous velocity of the mass.
 - (a) Find the equation of motion if it is released from a position 20 ft below its equilibrium position with a downward velocity of 41 ft/sec.
 - (b) Graph the solution and determine whether the motion is overdamped, critically damped, or underdamped.
9. A 9-kg mass is attached to a vertical spring with a spring constant of 16 N/m. The system is immersed in a medium that imparts a damping force equal to 24 times the instantaneous velocity of the mass.
 - (a) Find the equation of motion if it is released from its equilibrium position with an upward velocity of 4 m/sec.
 - (b) Graph the solution and determine whether the motion is overdamped, critically damped, or underdamped.
10. A 1-kg mass stretches a spring 61.25 cm. The resistance in the spring-mass system is equal to eight times the instantaneous velocity of the mass.
 - (a) Find the equation of motion if the mass is released from a position 5 m below its equilibrium position with an upward velocity of 10 m/sec.
 - (b) Determine whether the motion is overdamped, critically damped, or underdamped.
11. A 32-lb weight (1 slug) stretches a vertical spring 128 in. The resistance in the spring-mass system is equal to four times the instantaneous velocity of the mass.
 - (a) Find the equation of motion if it is released from its equilibrium position with a downward velocity of 12 ft/sec.
 - (b) Determine whether the motion is overdamped, critically damped, or underdamped.
12. A 64-lb weight is attached to a vertical spring with a spring constant of 4.625 lb/ft. The resistance in the spring-mass system is equal to the instantaneous velocity. The weight is set in motion from a position 1 ft below its equilibrium position with an upward velocity of 2 ft/sec. Is the mass above or below the equilibrium position at the end of π sec? By what distance?
13. A mass that weighs 8 lb stretches a spring 6 inches. The system is acted on by an external force of $8\sin(8t)$ lb. If the mass is pulled down 3 inches and then released, determine the position of the mass at any time.
14. A mass that weighs 6 lb stretches a spring 3 in. The system is acted on by an external force of $8\sin(4t)$ lb. If the mass is pulled down 1 inch and then released, determine the position of the mass at any time.

15. Find the charge on the capacitor in an RLC series circuit where $L = 40 \text{ H}$, $R = 30 \Omega$, $C = 1/200 \text{ F}$, and $E(t) = 200 \text{ V}$. Assume the initial charge on the capacitor is 7 C and the initial current is 0 A .
16. Find the charge on the capacitor in an RLC series circuit where $L = 2 \text{ H}$, $R = 24 \Omega$, $C = 0.005 \text{ F}$, and $E(t) = 12 \sin(10t) \text{ V}$. Assume the initial charge on the capacitor is 0.001 C and the initial current is 0 A .
17. A series circuit consists of a device where $L = 1 \text{ H}$, $R = 20 \Omega$, $C = 0.002 \text{ F}$, and $E(t) = 12 \text{ V}$. If the initial charge and current are both zero, find the charge and current at time t .
18. A series circuit consists of a device where $L = \frac{1}{2} \text{ H}$, $R = 10 \Omega$, $C = \frac{1}{50} \text{ F}$, and $E(t) = 250 \text{ V}$. If the initial charge on the capacitor is 0 C and the initial current is 18 A , find the charge and current at time t .